

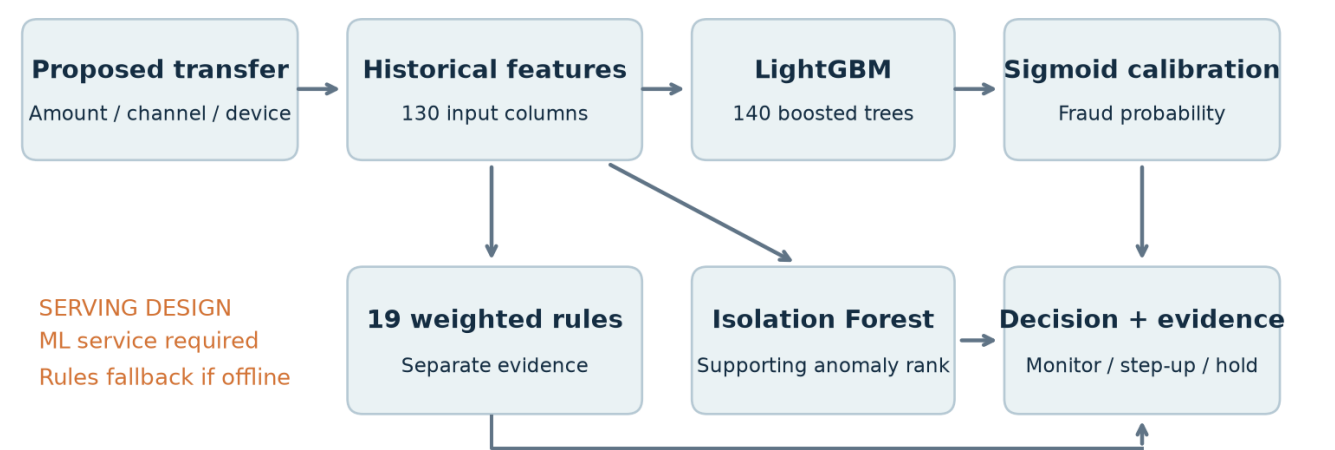
Fraud detection. Evidence before complexity.

Model findings, comparisons and technical implementation | 07 October 2026

Selected exported model: LightGBM_200k-v1

A compact, calibrated fraud-ranking model supported by interpretable rules and behavioral signals. The latest saved selection uses LightGBM from the 200k experiment, with sigmoid calibration and 130 input features.

0.2417 PR-AUC	10.90 ms P95	4.10% flagged
Common synthetic test	Prepared-frame, batch 1	At selected threshold



Scope: 100,000 fresh synthetic test transactions; 5,223 fraud cases. These are simulator findings, not real customer or upay production results. ML serving is configurable; when unavailable, the app falls back to rules. Benchmark results measure the saved model, not the complete app policy.

Evidence: backend/models/final/model_metadata.json; results/final_common_test/model_comparison.csv; src/lib/engine.ts (repository root).

Which model do we use?

Use the latest exported LightGBM model for the ML prototype; keep the evidence and operational policy separate.

Decision factor	LightGBM 200k	Hybrid without sequence 500k
Common-test PR-AUC	0.24168	0.24153
95% paired AP difference	Reference	-0.00624 to +0.00510 vs LightGBM
Batch-1 P95	10.90 ms	21.45 ms
Saved experiment bundle	0.113 MiB	2.037 MiB
At selected thresholds: F1	0.3255	0.3307
At selected thresholds: recall	29.06%	30.17%

Why this selection is defensible

LightGBM achieves almost the same ranking quality as the strongest hybrid with about half its measured P95 inference time and an approximately 18 times smaller saved bundle. The near-zero AP difference does **not** establish statistically superior detection. The hybrid has slightly higher F1 and recall at its own validation-selected threshold.

Resolve the older recommendation

The earlier per-scale report recommended 100k Hybrid_WithoutGraph using a validation-AP tolerance rule. The later common-test selection file and exported metadata name LightGBM_200k. This report uses the later exported artifact as the answer to “which model we use,” while retaining earlier findings as separate experiments.

Selection discipline

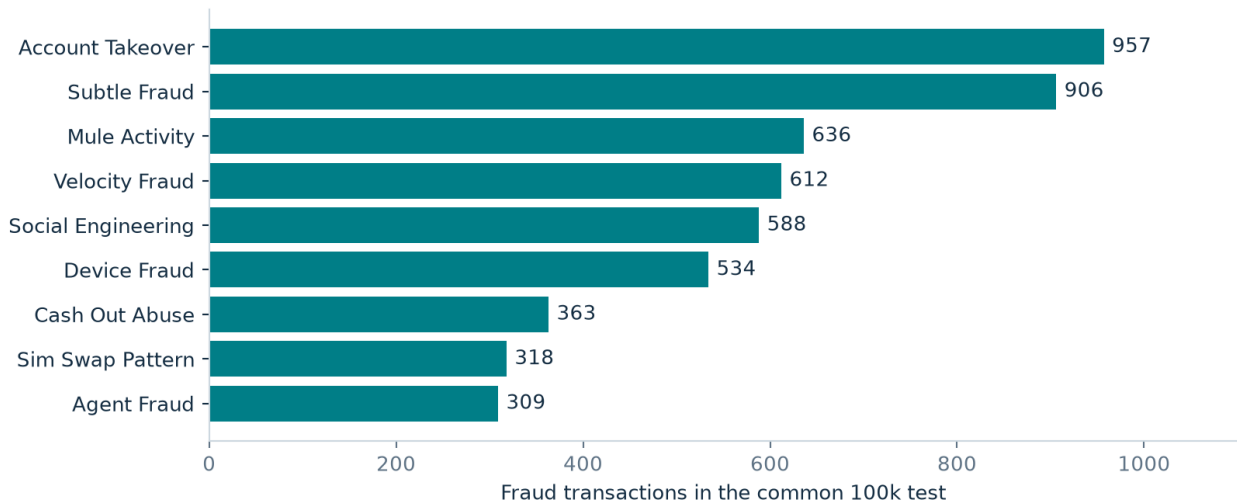
Final thresholds were selected on each training run's validation block. The repository does not establish that the final model choice was frozen before viewing common-test results. Treat the common set as comparative evidence; reserve another untouched holdout for confirmation after freezing the selection.

Evidence: final_common_test/final_selection.json, confidence_intervals.csv, model_comparison.csv; reports/final_report.md. All paths in this report are relative to mfs-guard-ml unless stated otherwise.

What did we evaluate?

Fresh simulated customers and episodes, drawn from the same generator as training.

Population	Common final test
Transactions / fraud	100,000 / 5,223 (5.223%)
Users / devices / agents	1,250 / 1,666 / 83
Replay / seed	120-day simulation / seed 2026
Cold starts / new devices	6.25% / 8.775% of transactions
Schema / inputs	154 columns / 130 model input columns
Transaction-ID overlap	0 with training datasets



The generator includes noisy outcomes, persistent wallets, multi-event fraud episodes, legitimate stress behavior and overlapping fraud signals. APP, AGENT, USSD, WEB and API channels are represented. All nine fraud categories occur in the common test.

Evidence: [reports/final_test_dataset_card.md](#); [src/data_generator.py](#); [results/final_common_test/fraud_type_performance.csv](#).

Chronological, separated evaluation

Five generated scales: 100k, 200k, 300k, 400k and 500k. Training seed 42; fresh common-test seed 2026.

200k run block	Share	Rows	Purpose
Train	70%	140,000	Fit preprocessing and base estimators
Tune	3%	6,000	Choose hyperparameters by AP
Fusion	3%	6,000	Fit logistic stacking model
Calibrate	3%	6,000	Fit calibration candidates
Select	3%	6,000	Choose calibration / system
Threshold	3%	6,000	Choose action operating points
Own temporal test	15%	30,000	Evaluate returning customers

Leakage controls

Features are emitted from prior history before state updates. Targets, fraud types, identifiers, post-decision balances, rule aggregates and simulator profile parameters are excluded from standalone supervised inputs. Imputation, scaling and categorical encoding are fitted on training rows only. Prefix-invariance and target-mutation tests are used to check causal history construction.

Two different evaluation questions

The per-scale 15% tests evaluate later transactions from returning customers (100% of 200k test users were seen in training). The common test replays fresh synthetic histories and includes the cold-start warm-up. Their absolute metrics should not be mixed or presented as one learning curve.

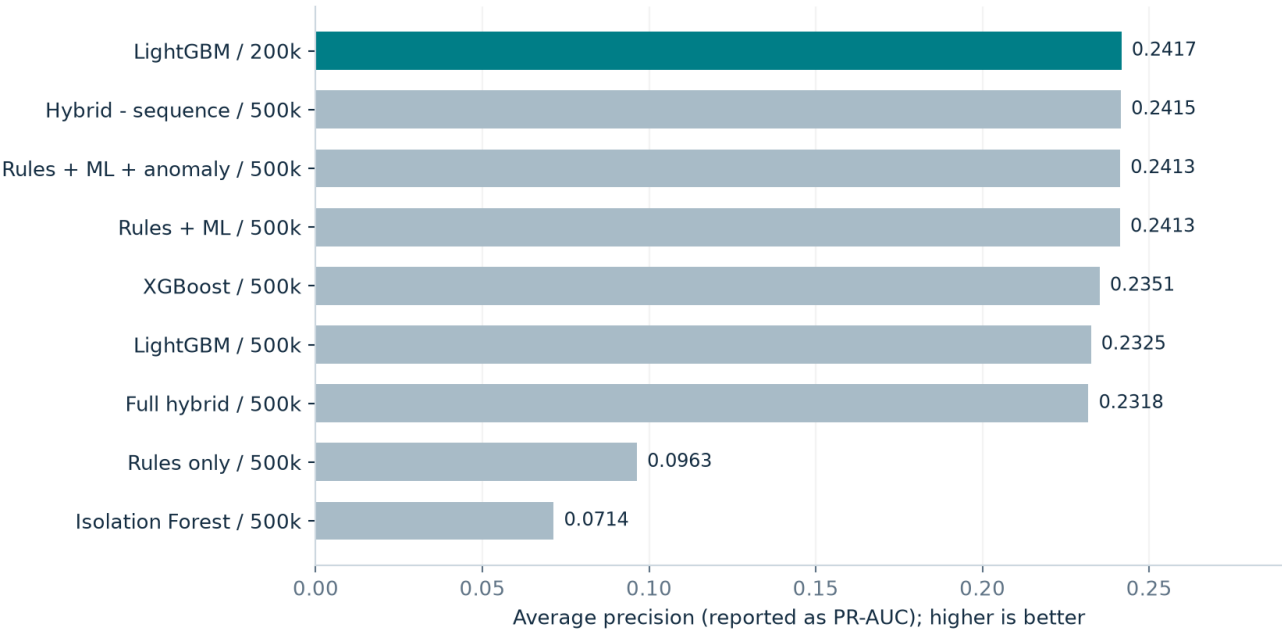
What is verified in saved evidence

The common-test dataset card records schema/accounting/chronology checks, 96 sampled history checks, a passed leakage audit, no missing cells and no transaction-ID overlap. These are saved findings; this report does not claim a new training or test-suite run.

Evidence: results/200k/split_boundaries.json; src/preprocessing.py; reports/final_test_dataset_card.md; reports/final_report.md.

Ranking quality across finalists

Every bar uses the same 100,000 synthetic transactions. Higher average precision is better.



The experiment labels average precision as “PR-AUC.” The prevalence reference is **0.05223**; LightGBM's AP is about **4.63 times** that value. This is a ranking-quality comparison, not a probability of catching all fraud.

LightGBM 200k (0.24168), Hybrid without sequence 500k (0.24153), Hybrid + anomaly 500k (0.24131) and Rules + ML 500k (0.24126) form a tight cluster. More layers do not automatically yield better ranking. Rules alone and Isolation Forest alone are substantially weaker here.

Selected model metric	Estimate	95% day-bootstrap interval
PR AUC	0.2417	0.2265 to 0.2577
PRECISION	0.3699	0.3505 to 0.3910
RECALL	0.2906	0.2721 to 0.3059
F1	0.3255	0.3085 to 0.3412

Evidence: final_common_test/model_comparison.csv; confidence_intervals.csv. 1,000 UTC-day bootstrap resamples, conditional on fitted models; no training-seed uncertainty.

The complete model scorecard

Precision, recall, F1 and flagged share use each model's own maximum-validation-F1 threshold.

Model / training scale	AP	ROC AUC	Precision	Recall	F1	Flagged
LightGBM / 200k	0.2417	0.7660	36.99%	29.06%	0.3255	4.10%
Hybrid - sequence / 500k	0.2415	0.7675	36.58%	30.17%	0.3307	4.31%
XGBoost / 500k	0.2351	0.7677	37.31%	28.05%	0.3202	3.93%
Full hybrid / 500k	0.2318	0.7667	35.69%	31.19%	0.3329	4.56%
Rules only / 500k	0.0963	0.6172	15.06%	22.59%	0.1807	7.84%
LightGBM / 500k	0.2325	0.7707	35.31%	32.80%	0.3401	4.85%
Rules + ML + anomaly / 500k	0.2413	0.7675	37.43%	28.49%	0.3235	3.98%
Rules + ML / 500k	0.2413	0.7674	37.36%	28.43%	0.3229	3.98%
Isolation Forest / 500k	0.0714	0.5780	9.07%	20.68%	0.1261	11.90%

How to read the metrics

Precision = caught fraud / all flagged transactions. **Recall** = caught fraud / all actual fraud. **F1** balances the two. **False-positive rate** = flagged legitimate / all legitimate. **AP** summarizes precision over recall as the threshold varies; **ROC AUC** measures positive-versus-negative ranking.

The selected model has MCC **0.2954**, balanced accuracy **0.6317**, false-positive rate **2.7285%**, false-negative rate **70.9362%** and Brier score **0.04399**. Accuracy alone is misleading: predicting every transaction as legitimate already yields 94.777% accuracy on this population.

Fair budget comparison

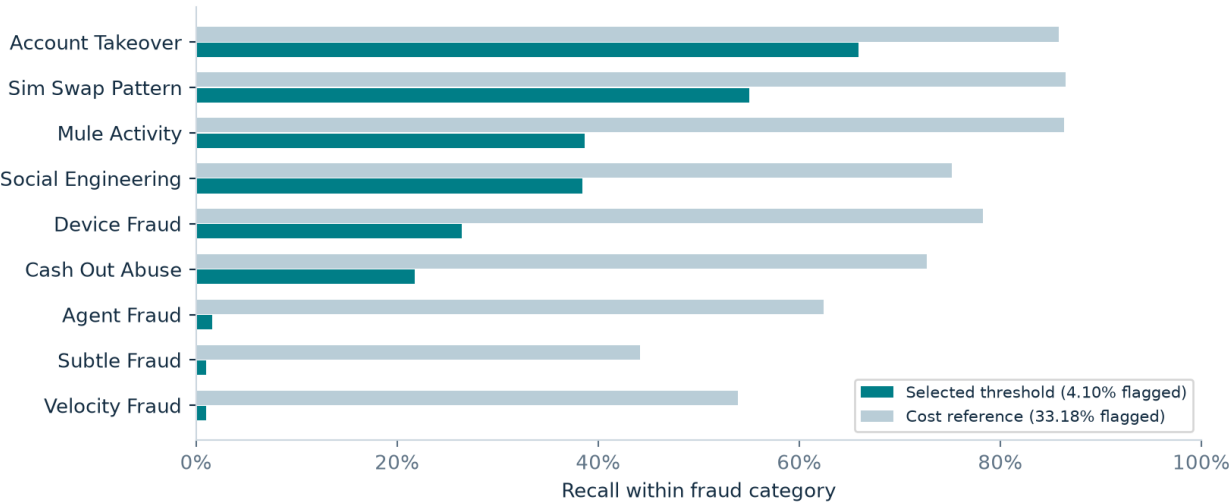
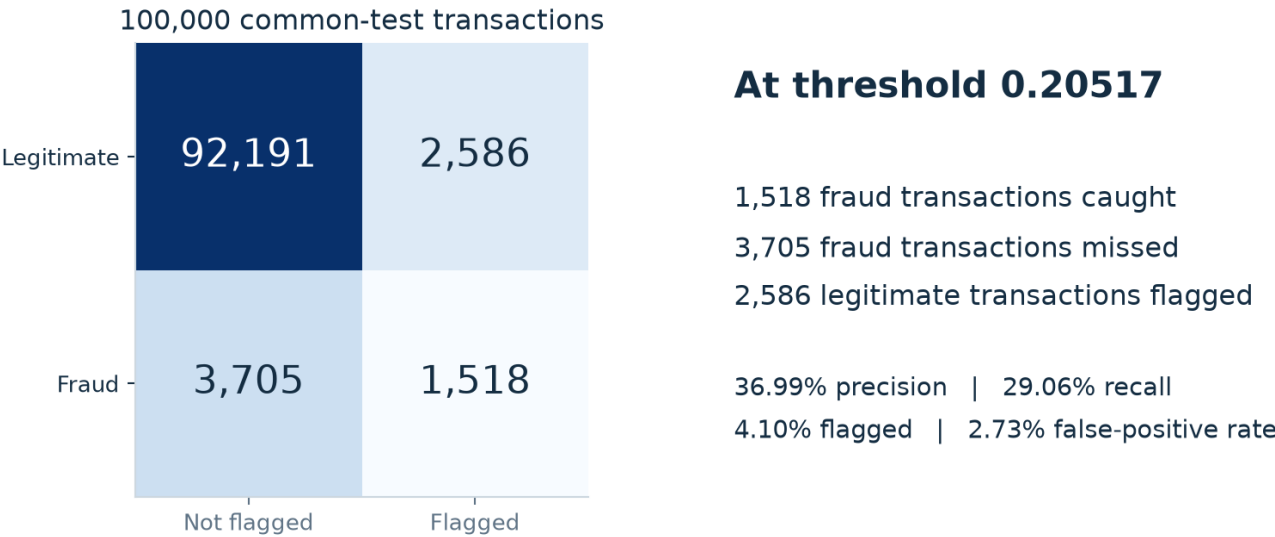
LightGBM ranked review budget	Recall	Precision
Top 1%	8.86%	46.30%
Top 2%	16.20%	42.30%
Top 5%	32.89%	34.36%

Top-budget results rank the test scores and use fractional random tie-breaking at the boundary. They describe queue capacity; they are not newly tuned deployable probability thresholds.

Evidence: final_common_test/model_comparison.csv; src/common_test.py (budget_recall).

What the chosen model misses

A useful judge-facing claim must include both detections and misses.

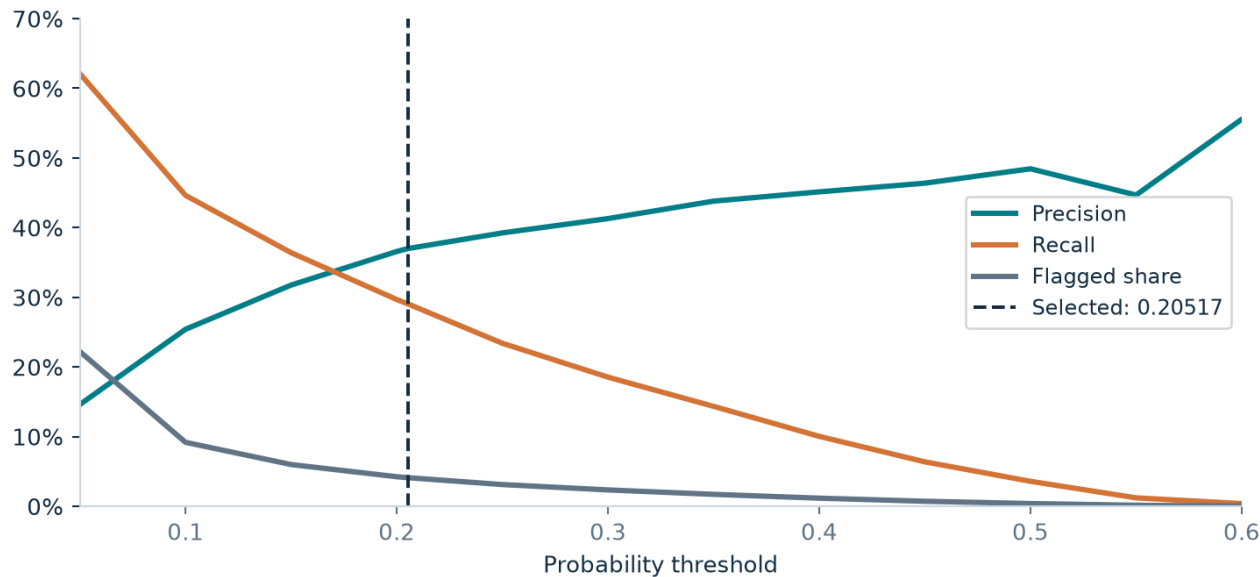


Main weakness: subtle fraud recall is 0.99%, velocity fraud 0.98% and agent fraud 1.62% at the selected threshold. Account takeover is stronger at 65.94%. Subtle fraud contributes 897 missed cases. The lower cost-reference threshold improves recall while flagging one-third of all transactions.

Evidence: final_common_test/model_comparison.csv; fraud_type_performance.csv. Matrix values are counts; cell shading uses a log scale for readability.

Probability is not the action

Calibrated probability, weighted rules and operational capacity are separate decisions.



Exported ML band	Threshold	Intended tier
Below monitor	$p < 0.082229$	Approve
Monitor	$0.082229 \leq p < 0.205167$	Approve and monitor
Step-up	$0.205167 \leq p < 0.445$	Additional authentication
Hold	$p \geq 0.445$	Temporary hold / review

The step-up threshold **0.20516681** maximizes F1 on the original validation threshold block. Monitor and hold are chosen from validation budgets of 15% and 1%. These budgets are not guarantees on future data. On the common test, 7,506 are monitor-only, 3,337 step-up and 767 hold.

The exported policy also records rule-score thresholds 20 / 40 / 70 and a reject condition requiring both ML at least hold and rule score at least 70. The benchmarked tiered policy measures monitor / step-up / hold only; it does not measure that combined reject rule. Sweeps above are descriptive and must not be used to retune on this test.

Evidence: backend/models/final/thresholds.json; final_common_test/threshold_comparison.csv, business_cost_comparison.csv; src/common_test.py.

Higher recall has a workload price

All costs are assumed simulation units. They are not measured savings or real provider economics.

Policy on common test	Flagged / interrupted	Recall	Estimated cost
Selected validation tiers	4,104 (4.10%)	29.06% at step-up	3,834,605
Lower binary cost reference	33,178 (33.18%)	70.69%	2,667,786
Approve everything (derived)	0	0%	5,223,000

Assumptions behind the figures

Each fraud case carries 1,000 loss units before intervention. Assumed residual fraud loss: approve 100%, monitor 80%, step-up 25%, hold 5%. Step-up costs 2 units per affected transaction; hold costs 8. Interrupted legitimate users incur 5 friction units; held legitimate users also incur 15 investigation units.

The selected tiered policy's 3,834,605 total consists of 3,802,700 residual fraud-loss units, 19,095 false-positive costs, 6,674 verification costs and 6,136 hold costs. The code does not add a separate monitor operating fee in this common-test calculation.

Avoid confusing two cost functions

The lower threshold 0.03734065 was selected using the training-run binary objective **1,000 x false negatives + 20 x false positives**. The common-test policy evaluation then applies the action-specific residual losses above. "Cost-optimal" is a reference to the original validation objective, not proof that it is globally optimal for the tiered policy.

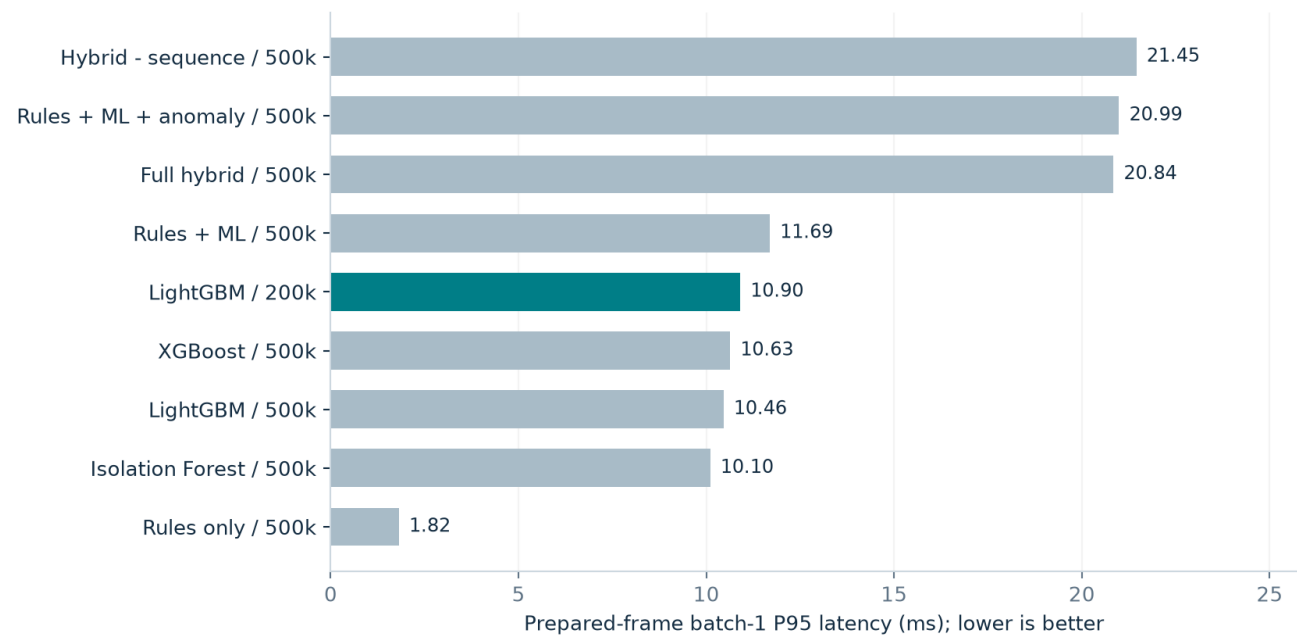
A defensible operating choice

The selected threshold controls interruption volume but misses most fraud. A real deployment would need actual fraud loss, verification efficacy, analyst capacity and customer friction estimates before choosing a policy. The lower threshold's 11.13% precision illustrates the cost of increased recall.

Evidence: config.yaml; src/common_test.py (business_cost); final_common_test/business_cost_comparison.csv and model_comparison.csv. Approve-all cost derived as 5,223 x 1,000.

Compact inference, measured honestly

Timing includes preprocessing, saved model dependencies and calibration on prepared historical feature frames.



Batch	Repeats	P50 ms	P95 ms	Transactions/s
1	300	8.47	10.90	112
10	100	8.31	9.89	1,173
100	40	8.75	10.54	11,181
1000	12	15.16	16.85	65,742

LightGBM has 140 trees, a 0.113 MiB saved experiment bundle, and about 7.80 MiB measured process-RSS growth after loading and scoring. Its training/search wall time was 5.49 seconds; sampled peak process RSS was 1,031.46 MiB. Those memory measures have different scopes.

Boundary: single process, four threads, round-robin benchmark. State retrieval, feature-history construction, HTTP and network delay are excluded. Exact benchmark hardware is not recorded in the cited comparison. These results do not establish production latency or online throughput.

Evidence: final_common_test/latency_comparison.csv, model_comparison.csv; models/200k/LightGBM_metadata.json; src/common_test.py.

Inside the selected LightGBM

200k names the full generated experiment; the base estimator was fitted on its 140k training rows.

Configuration	Saved value
Algorithm	Gradient-boosted decision trees (gbdt)
Trees / leaves / learning rate	140 / 15 / 0.06
Regularization / min child samples	L2 (reg_lambda) = 8 / 30
Class imbalance	class_weight = balanced
Sampling	subsample = 1.0; colsample_bytree = 1.0
Execution	4 threads; random_state 42; deterministic; force_col_wise
Tuning	2 seeded trials: reg_lambda 3 vs 8, both 15 leaves
Winning tune-block AP	0.24661649
Calibration	Sigmoid, fitted on a separate 6k block

Train-only preprocessing

Numeric columns: median imputation followed by standard scaling. Categorical columns: most-frequent imputation and dense one-hot encoding with unknown values ignored. The bundle receives 130 engineered/raw input columns; the post-encoding dimension can be larger.

Feature families

Transaction and account context; amount relative to historical behavior; 5-minute to multi-day velocity windows; device and recipient novelty; cash-flow and turnaround behavior; incremental graph connectivity; location and travel; hour/channel deviation; credential and transfer sequences; agent activity. Historical state must match the offline computation at serving time.

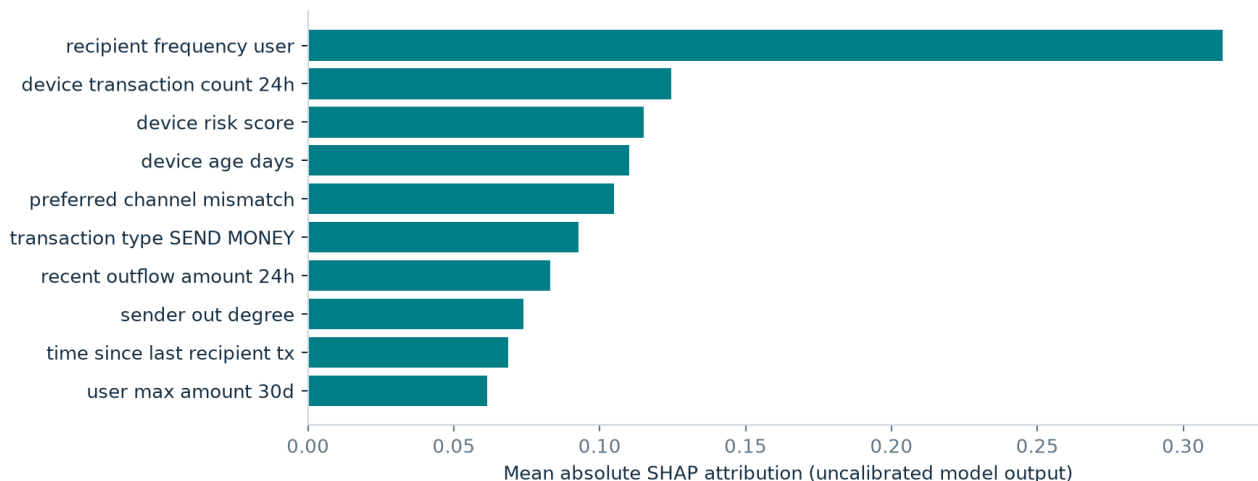
Calibration methodology

Identity, sigmoid and isotonic candidates are evaluated using separate fitting and selection blocks. Selection minimizes Brier score subject to an AP-loss limit of 0.015. The exported model uses sigmoid; probability calibration is still only validated within the simulator distribution.

Evidence: models/200k/LightGBM_metadata.json; src/preprocessing.py; reports/final_report.md; backend/models/final/feature_list.json.

Which inputs influence predictions?

Global SHAP attributions for the selected LightGBM from the earlier 200k experiment.



Recipient familiarity is the strongest feature in this sampled explanation. Device activity and age, device risk, preferred-channel mismatch, outgoing cash flow and graph degree also contribute. This supports a multi-signal explanation, rather than a single amount threshold.

Explainability boundaries

The saved analysis uses 256 randomly sampled held-out transactions for global SHAP and the 12 highest-risk examples for local explanations. These attributions describe the uncalibrated tree model, not the final calibrated probability, causal drivers, or the entire action policy. This is not a SHAP rerun on the common final test.

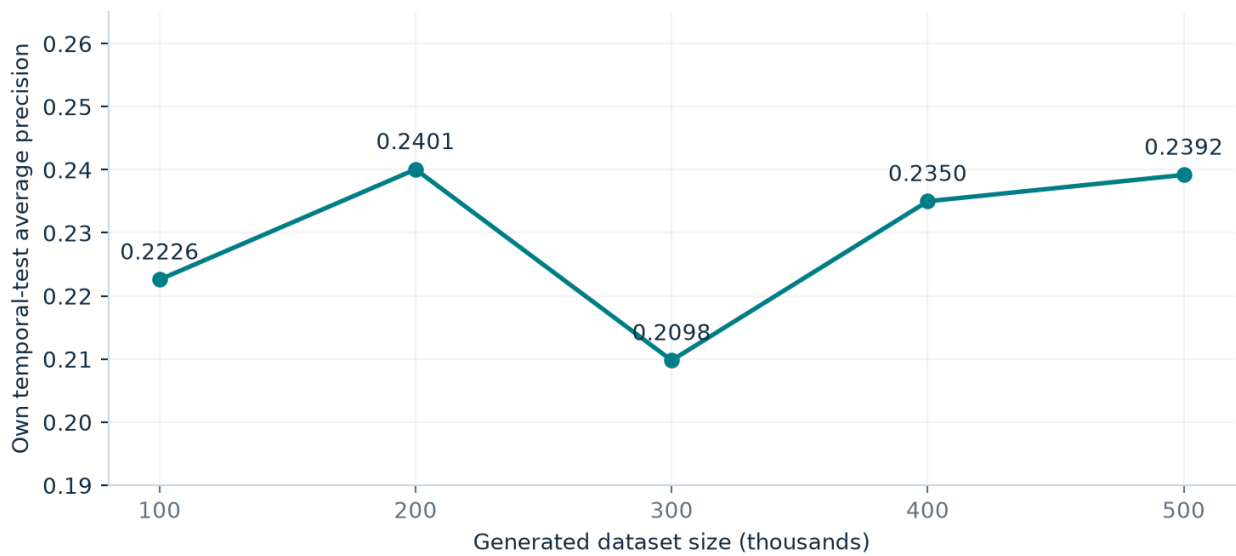
Evidence judges can inspect

The repository includes global SHAP values, native feature importance, local explanations, fraud-type error slices and weighted rule evidence. Keep rule reasons distinguishable from learned-model attributions: a triggered rule does not prove that a particular tree caused the decision.

Evidence: results/200k/shap_feature_importance.csv, explainability_scope.json, local_explanations.json. Figure redrawn from saved numeric SHAP values.

Does more data or complexity help?

Earlier per-scale findings are informative, but evaluate different populations and different selected systems.



Scale	Validation-selected system	Own-test AP
100k	Hybrid_WithoutGraph	0.2226
200k	LightGBM	0.2401
300k	LightGBM	0.2098
400k	LightGBM	0.2350
500k	Hybrid_AddAnomaly	0.2392

The scale curve is non-monotonic. It grows population size while keeping approximate per-user history depth fixed; it is not a fixed-population learning curve and uses only one training/generation seed.

500k ablation finding	AP effect	95% interval
Full hybrid minus no-anomaly fusion	+0.0043	+0.0015 to +0.0070
Supervised minus graph-removed retraining	+0.0018	-0.0009 to +0.0047
Supervised minus sequence-removed retraining	+0.0055	+0.0018 to +0.0100

These earlier paired day-bootstrap intervals are conditional on fitted models (150 repeats) and not adjusted for multiple comparisons. Full hybrid was 0.0020 AP below standalone XGBoost on that own-test comparison. Correlated features can preserve proxies for removed families.

Evidence: results/final/dataset_size_comparison.csv; reports/final_report.md; results/final/ablation_all_sizes.csv. Not common-final-test ablations.

What exists, and what remains?

The workspace now contains a Python scoring service and a configurable Next.js connection, with a rules fallback.

Repository layer	Observed implementation
ML experiment	Saved models, calibrators, thresholds, dataset cards, comparisons and error analyses
Final exported bundle	LightGBM model, preprocessor, calibrator, feature list, metadata, policy thresholds and config snapshot
Supporting anomaly export	500k Isolation Forest component; separate from selected LightGBM probability
Next.js ML connection	src/lib/ml.ts calls the Python scoring API when MFS_ML_SERVICE_URL is configured
Action integration	The backend takes the more severe ML/rule action; combined high scores or blacklists can reject. Next.js preserves hard rule blocks.
Failure fallback	Missing service configuration or scoring failure returns DEGRADED and retains the local rule decision

Serving contract in the current code

The Python feature service reuses offline feature engineering on a chronologically ordered replay of stored history plus the current transaction. The saved preprocessor, LightGBM and sigmoid calibrator produce probability. The decision service combines ML and rule bands; anomaly scores and SHAP factors provide supporting evidence. The Next.js client uses the returned action, with local hard blocks retained.

Remaining serving limitations

The feature service replays a capped history ledger for each request, so feature work grows with history length and truncation can change long-history or lifetime graph features. Production needs incremental state, offline/online parity checks and full-request latency measurement. This report inspected integration code but did not verify a running configured service or deployment.

Do not label the current app's 0-100 weighted rule score as the LightGBM fraud probability. The benchmark's model metrics do not measure the present app's decisions.

Evidence: backend/models/final/*; scripts/export_final_model.py; backend/app/services/feature_engine.py and decision_engine.py; src/lib/ml.ts, src/lib/engine.ts and src/app/api/v1/[...path]/route.ts (repository root).

Key findings and honest limitations

Suggested presentation narrative and technical answers.

Our model choice

“We exported calibrated LightGBM from the 200k experiment. On a fresh 100k synthetic comparison, it delivers 0.2417 average precision with 10.90 ms prepared-frame P95 inference. A much larger hybrid gives essentially the same ranking quality, so we prefer the simpler model.”

Our operational trade-off

“At the validation-selected threshold, we flag 4.1% of transactions, with 37.0% precision and 29.1% recall. A lower reference threshold catches 70.7% of fraud but flags 33.2% of transactions. We expose this trade-off instead of claiming a single universal threshold.”

What the prototype demonstrates

A reproducible synthetic benchmark; separated chronological fitting and evaluation; learned ranking compared against rules and anomaly baselines; saved calibration and threshold artifacts; uncertainty estimates; and actionable diagnostics across fraud types and workload budgets.

What it does not establish

Real-world accuracy, real provider cost savings, demographic fairness, adaptive-attacker robustness, cross-generator transfer, or production serving latency. One generator family and one training seed limit generalization. The largest missed category is subtle fraud; known-device and known-recipient behavior remain difficult.

Next experiment

Freeze the chosen bundle and policy, then evaluate on another untouched holdout. Repeat training seeds, include user-disjoint and temporal stress tests, estimate real costs, validate on consented de-identified event histories with delayed labels, and monitor analyst overrides and drift during shadow deployment.

Evidence: Synthesis of the cited local experiment artifacts. Recommendation is for a synthetic ML prototype, not autonomous live customer decisions.

All 130 model inputs

Saved preprocessor input order. IDs, target labels and post-decision outcomes are not included.

001 transaction_type	002 channel
003 currency	004 amount
005 customer_segment	006 account_age_days
007 failed_pin_attempts	008 otp_resend_count
009 balance_inquiry_count_5m	010 pin_reset_recently
011 password_reset_recently	012 biometric_failed_recently
013 rooted_device	014 emulator_detected
015 vpn_active	016 screen_share_detected
017 sim_changed_recently	018 device_os_changed
019 device_fingerprint_changed	020 blacklisted_device
021 blacklisted_agent	022 latitude
023 longitude	024 fee
025 balance_before	026 depletion_ratio
027 behavioral_history_count	028 user_mean_amount
029 user_median_amount	030 user_std_amount
031 amount_vs_user_mean	032 amount_vs_user_median
033 amount_zscore_user	034 amount_percentile_user
035 time_since_previous_tx_seconds	036 behavioral_model_active
037 tx_count_5m	038 amount_sum_5m
039 tx_count_15m	040 amount_sum_15m
041 tx_count_1h	042 amount_sum_1h
043 avg_amount_1h	044 tx_count_6h
045 amount_sum_6h	046 tx_count_24h
047 amount_sum_24h	048 avg_amount_24h
049 max_amount_24h	050 user_max_amount_30d
051 user_mean_daily_tx_count	052 user_mean_hourly_tx_count
053 device_is_new	054 device_age_days
055 device_first_seen	056 shared_device_user_count
057 user_device_count	058 device_transaction_count_24h
059 device_transaction_count_7d	060 channel_changed_recently
061 preferred_channel_mismatch	062 first_time_recipient
063 recipient_frequency_user	064 user_receiver_tx_count_7d
065 user_receiver_tx_count_30d	066 time_since_last_recipient_tx

Evidence: backend/models/final/feature_list.json; 130 exact input names. Categorical one-hot expansion occurs inside the saved preprocessor.

Model inputs, continued

Saved preprocessor input order. IDs, target labels and post-decision outcomes are not included.

067 receiver_transaction_count_24h	068 receiver_unique_senders_24h
069 receiver_transaction_count_7d	070 receiver_unique_senders_7d
071 recent_inflow_amount_1h	072 recent_outflow_amount_1h
073 receiver_inflow_1h	074 receiver_outflow_1h
075 recent_inflow_amount_24h	076 recent_outflow_amount_24h
077 receiver_inflow_24h	078 receiver_outflow_24h
079 turnaround_latency_seconds	080 rapid_fund_turnaround
081 receiver_turnaround_ratio	082 receiver_sender_diversity
083 sender_in_degree	084 sender_out_degree
085 receiver_in_degree	086 receiver_out_degree
087 unique_sender_count	088 unique_receiver_count
089 connected_component_size	090 reciprocal_transfer_ratio
091 shared_recipient_count	092 fund_concentration
093 rapid_forwarding_ratio	094 shared_device_count
095 mule_network_score	096 recipient_risk_score
097 device_risk_score	098 previous_latitude
099 previous_longitude	100 distance_from_previous_tx_km
101 time_since_previous_location_seconds	102 distance_from_home_km
103 impossible_travel_speed	104 impossible_travel
105 transaction_hour	106 day_of_week
107 weekend_flag	108 is_night_transaction
109 unusual_hour_score	110 unusual_channel_score
111 unusual_location_score	112 unusual_transaction_type_score
113 user_behavior_score	114 pin_reset_then_transfer
115 new_device_then_transfer	116 failed_pin_then_success
117 balance_inquiry_then_cashout	118 incoming_then_rapid_outgoing
119 new_recipient_then_large_transfer	120 multiple_otp_then_transfer
121 device_change_then_high_value	122 sim_change_then_transfer
123 credential_change_then_cashout	124 channel_hop_sequence
125 sequence_risk_score	126 agent_transaction_count_1h
127 agent_cashout_amount_1h	128 agent_transaction_count_24h
129 agent_cashout_amount_24h	130 agent_risk_score

Evidence: backend/models/final/feature_list.json; 130 exact input names. Categorical one-hot expansion occurs inside the saved preprocessor.

All models across all five scales

Average precision on each scale's own temporal test. Compare within columns; populations differ across columns.

System	100k	200k	300k	400k	500k
FullHybrid	0.2235	0.2298	0.2141	0.2357	0.2318
HistGradientBoosting	0.2327	0.2110	0.2057	0.2337	0.2168
Hybrid_AddAnomaly	0.2295	0.2348	0.2082	0.2245	0.2392
Hybrid_AddDeviceRecipient	0.2303	0.2335	0.2113	0.2225	0.2324
Hybrid_RulesSupervised	0.2303	0.2360	0.2101	0.2246	0.2388
Hybrid_WithoutAnomaly	0.2234	0.2308	0.2152	0.2360	0.2275
Hybrid_WithoutGraph	0.2226	0.2307	0.2140	0.2207	0.2283
Hybrid_WithoutSequence	0.2300	0.2325	0.2113	0.2224	0.2421
IsolationForest	0.0634	0.0785	0.0681	0.0749	0.0707
LightGBM	0.2317	0.2401	0.2098	0.2350	0.2305
LogisticRegression	0.2049	0.2047	0.1866	0.2187	0.2197
RandomForest	0.2108	0.2218	0.1947	0.2161	0.2267
RulesOnly	0.0850	0.0888	0.0781	0.0863	0.0924
WithoutGraphFeatures	0.2258	0.2392	0.2113	0.2361	0.2319
WithoutSequenceFeatures	0.2307	0.2335	0.2074	0.2314	0.2282
XGBoost	0.2297	0.2293	0.2076	0.2248	0.2337

The study evaluates logistic regression, random forest, histogram gradient boosting, XGBoost, LightGBM, rules, Isolation Forest, progressive hybrid fusion, leave-one-component-out fusion and retrained feature-family removals. Hybrid fusion uses a logistic model fitted on its dedicated validation block.

This table supplies the broader research comparison; it must not replace the common-test model selection table. AP is threshold-independent, while the older report's high-recall operating points use a different cost-selected policy.

Evidence: results/final/all_model_results.csv; src/train_supervised.py; reports/final_report.md.

Policy details and source manifest

Report created from saved local evidence. No model was retrained and no test outcome was fabricated.

Rule feature	Threshold	Weight
depletion_ratio	0.8	12
depletion_ratio	0.95	10
failed_pin_attempts	3	12
otp_resend_count	3	8
pin_reset_recently	1	5
device_is_new	1	4
new_device_then_transfer	1	8
rooted_device	1	6
emulator_detected	1	10
vpn_active	1	3
tx_count_5m	3	12
rapid_fund_turnaround	1	10
channel_hop_sequence	1	8
balance_inquiry_count_5m	3	8
impossible_travel	1	10
recipient_risk_score	70	10
blacklisted_device	1	20
blacklisted_agent	1	20
sim_change_then_transfer	1	10

These are the 19 exported research rules. Rule-score bands are 20 / 40 / 70. Do not substitute this table for the current web app's independently configured rule policy.

Evidence: backend/models/final/thresholds.json; config.yaml. See the next page for source provenance and artifact hashes.

Reproduce the evidence

Primary evidence hierarchy: latest final selection and exported metadata, common-test tables, then earlier per-scale findings.

Evidence	Local path under mfs-guard-ml/
Final selection / version	results/final_common_test/final_selection.json; backend/models/final/model_metadata.json
Common-test scores / uncertainty	results/final_common_test/model_comparison.csv; confidence_intervals.csv
Policy / cost / latency	results/final_common_test/threshold_comparison.csv; business_cost_comparison.csv; latency_comparison.csv
Population / errors	reports/final_test_dataset_card.md; results/final_common_test/fraud_type_performance.csv; error_analysis.csv
Training / preprocessing	models/200k/LightGBM_metadata.json; src/train_supervised.py; src/preprocessing.py
Explanations / scale study	results/200k/shap_feature_importance.csv; results/final/dataset_size_comparison.csv; all_model_results.csv
Exports / serving boundary	scripts/export_final_model.py; backend/app/services/; backend/models/final/; repository-root src/lib/ml.ts and src/lib/engine.ts

SHA-256 of the selected exported components

model.joblib

586a69688cb466f823a25603a55be306
d051de592afff3ec3aa0a711461ca23a

preprocessor.joblib

80added98c4e2861085e32a227d7a3ea
6f3752a80fa1ddfa94de9660194a78cd

calibrator.joblib

ab0ba9e3e260931873bdb5d50a9bc866
0897a8937fd9ef161853e32779a74327

feature_list.json

6c5bdbc6990f08ea3f850542f8fad950
57c5fcd6b0433cf741f0597e61999ef5

thresholds.json

0ead3e82c9ec34058948e8f9819ec359
48fc0ab0087f3edc12e36bd2f94e1bd6

Figures were redrawn from saved numeric artifacts; the architecture image is a technical illustration. The report builder is tmp/pdfs/build_report.py at the repository root. Use the saved joblib bundles only in a compatible Python environment; preserve the feature list, preprocessing, calibration and versioned policy together.